Video Transcript

Video 6: Using DASK to Read and Analyze Multiple Files in Parallel (05:41)

So, we saw how to use DASK to create 30 files in parallel. Each of these files was 86,401 lines long. And if we look at the columns, we have timestamp, id, name, X, and Y. So, now let's see how we can read these. So, here was our code for writing them and we wrote them in parallel. And so remember they all start with 2000. So, let's take a look at reading them. So, here we're going to import DASK dataframe as DDF. Import DASK delayed, we don't need to import that.

And so, here we're going to read in parallel. So, we do ddf.read_CSV and we give a wild card; the star. So, any of the files starting 2000*. So, there should be 30 of those, and then we can print the head.

So, let's take a look at running that first. So, here, let me clear this and we'll run Python desk 04. So, you'll see here that it's printed out the head. Now, the head of what? What is it read in? Let's take a quick look. So, what it's read in, is a DASK dataframe. And a dataframe consists of multiple Pandas dataframes.

So, each of our dated files is going to be a Pandas dataframe. Now this whole area, you'll need to dig in a lot deeper, but the DASK reads and writes are very similar to the Pandas reads and writes. So, let's go back to our code. Let's now do some calculations on what we've read in. So, let's take a look now at printing out, for example, let's get the mean of the X column. So, first we need to do a df.compute to make sure that we've actually got a dataframe. Remember, this is a DASK one.

Now we can do, for example, we know there's an X column, so that'll give us the X column and we could calculate the mean and we need to do compute. Now, let's get the result mean and print it out. Okay, that looks good. Let's save it and let's run it. So, notice now it's taking a little while, but it's calculated the mean. And we can similarly calculate the mean of the Y column. Let's take a look how, just to make sure, what our data structure looks like now. Let's take a look at the number of columns.

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So, we can do df.columns and print that out. And let's do the rows as well. So, let me, I'm just going to copy and paste this and instead of columns, we'll have rows and here instead of that we need the index because that'll give us the rows. Okay, let's take a look and run it. So, we've got five columns and we've got 2.59 million rows. So, you remember that we had around 86,000 rows in each of those 30. If you do the multiplication, you'll come up with 2.5 million. So, we're handling fairly large data files here, certainly in the millions. So, that's reading using DASK and DASK's parallel computing powers. So, one, you repeat what I've done here so that you understand just how this works. OK, bye for now.

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